CLINICAL · APPROACH TO

Approach to hearing loss

Almost every patient with hearing loss can be placed with three questions: is the loss conductive or sensorineural, is it in one ear or both, and how quickly did it come on? The answers point to the cause and to what must not be missed. A one-sided or sudden sensorineural loss always needs more than a hearing aid.

AT A GLANCE

- Sort by type, side and speed.
- Forks at the bedside; audiogram to quantify.
- Asymmetric or sudden sensorineural loss needs urgent work-up.
- Treat reversible causes, then rehabilitate.

1 CHAPTER ONE Overview

A method, not a list.

1.1 Why a method

The causes of hearing loss run into hundreds. A structured approach, starting with tuning forks and an audiogram, narrows them quickly and catches the few that are urgent or dangerous.

1.2 The three questions

THREE QUESTIONS

- TYPE **Conductive or sensorineural?** or mixed, or non-organic
- SIDE **One ear or both?** asymmetry needs imaging
- SPEED **Sudden, progressive or fluctuating?** sudden loss is an emergency

2 CHAPTER TWO Classification

Where the lesion is, and how bad.

2.1 By site

2.1.1 Conductive

Any lesion from the canal to the stapediovestibular joint that stops sound reaching the cochlea.

2.1.2 Sensorineural

A lesion of the cochlea (*sensory*), the cochlear nerve (*neural*), or the central pathways.

2.1.3 Mixed

Both together, as in otosclerosis with cochlear involvement or chronic otitis media with labyrinthine damage.

2.1.4 Non-organic

Claimed loss without an organic basis, deliberate or unconscious.

2.2 By degree

WHO GRADES OF HEARING LOSS (2021)

GRADE	BETTER-EAR AVERAGE AT 500, 1000, 2000 AND 4000 HZ
Normal	Below 20 dB
Mild	20 to under 35 dB
Moderate	35 to under 50 dB
Moderately severe	50 to under 65 dB
Severe	65 to under 80 dB
Profound	80 to under 95 dB
Complete or total	95 dB or more

WHERE THE BOOKS DIFFER · GRADING SYSTEMS

Dhingra

Uses the older WHO scale averaged over 500, 1000 and 2000 Hz, with 0 to 25 dB counted as normal and 91 dB or more as profound.

The 2021 World Report on Hearing lowered the threshold for normal hearing to under 20 dB, added 4000 Hz to the average and added a complete hearing loss grade, because even mild losses affect communication. Many British services use the British Society of Audiology bands, in which normal hearing is up to 20 dB.

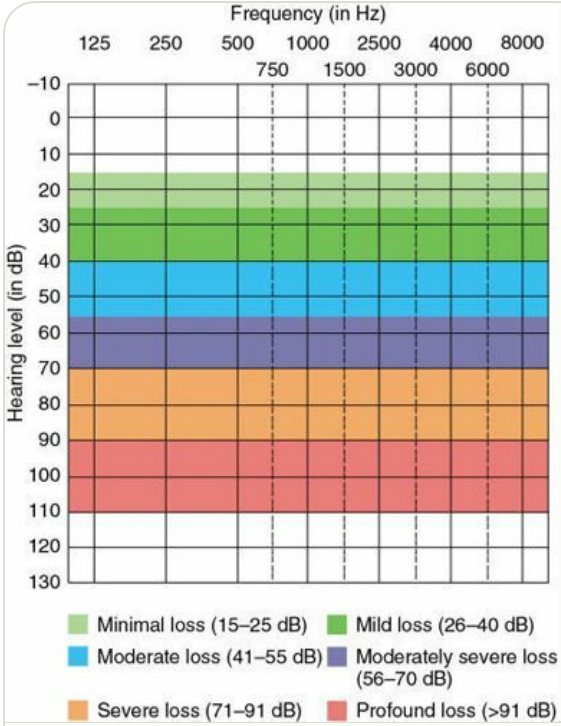


FIGURE 1 Classification of hearing loss by threshold. Most people with normal hearing have thresholds between −10 and +10 dB.

Dhingra, fig. 5.8

3 CHAPTER THREE Conductive loss

Sound that cannot get in.

3.1 Causes

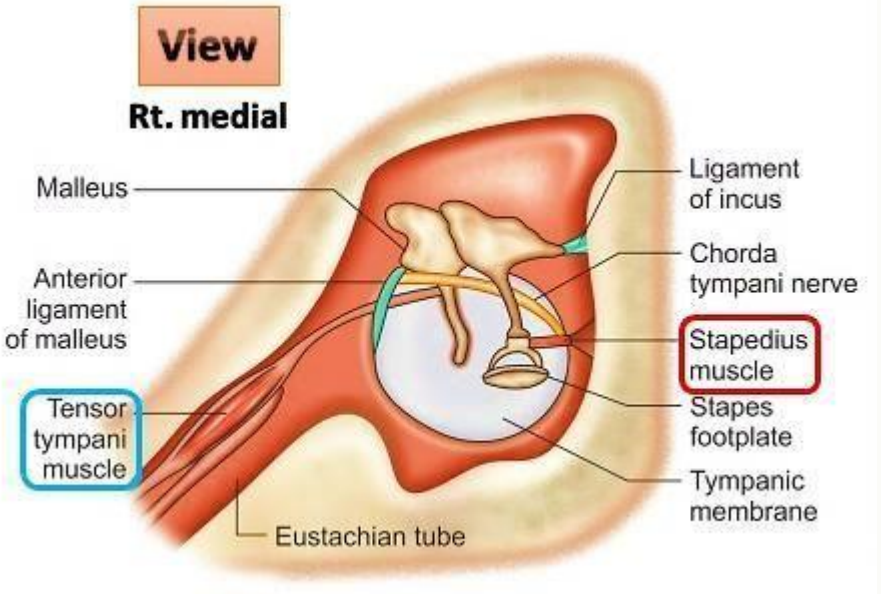


FIGURE 2 The drum, ossicular chain and intratympanic muscles: every conductive loss interrupts this chain somewhere.

Anatomy atlas · ENT Steps p. 51

3.1.1 Congenital

Canal atresia, fixation of the stapes or malleus head, ossicular discontinuity, and congenital cholesteatoma.

3.1.2 Acquired

- **Canal:** wax, foreign body, otitis externa, stenosis, tumours.
- **Drum:** perforation, scarring.
- **Middle ear:** fluid, blood, pus, mass, cholesteatoma, tympanosclerosis.
- **Ossicles:** disruption (trauma, erosion) or fixation (otosclerosis, tympanosclerosis).
- **Eustachian tube** failure.

3.2 Features

Low tones are affected more; speech discrimination is good; the patient may hear better in noise; voice is soft. A conductive loss alone never exceeds about 60 dB.

HOW BIG IS THE GAP?

- Complete canal block: about 30 dB.
- Perforation: 10 to 40 dB, more for larger and posterior holes.
- Malleus fixation: 10 to 25 dB.
- Ossicular discontinuity behind an intact drum: about 54 dB; with a perforation, about 38 dB, because an intact drum blocks sound reaching the round window.
- Closed oval window: about 60 dB.

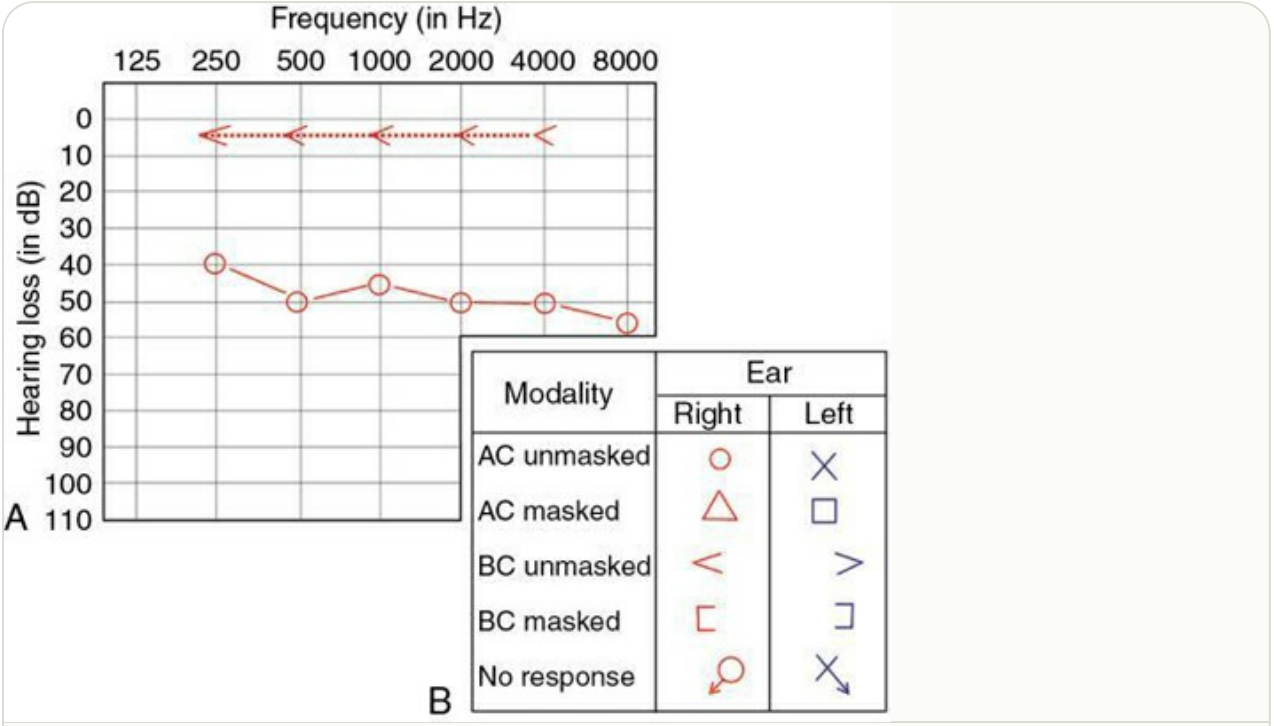


FIGURE 3 Audiogram of a right ear with a conductive loss and an air-bone gap, with the standard audiogram symbols.

Dhingra, fig. 5.1

3.3 Conductive loss behind an intact drum

When the drum looks normal, the list is **otosclerosis** (the commonest, with a family history, onset in early adult life, worsening in pregnancy, a Carhart notch at 2 kHz and absent stapedial reflexes), **ossicular discontinuity** after trauma, **malleus or**

incus fixation, **tympanosclerosis**, **congenital cholesteatoma** (a white mass behind an intact drum in a child), a small **effusion**, and **congenital ossicular anomalies**. A **third-window lesion** such as superior canal dehiscence produces an air–bone gap at low frequencies that is not truly conductive: bone conduction is better than normal, and the stapedial reflexes are present. Present reflexes with an apparent conductive loss should therefore stop a surgeon from operating on the stapes until a CT has excluded dehiscence.

3.4 Management

Clear the canal, drain fluid, remove a mass, repair the drum and chain (tympanoplasty), stapes surgery, or a hearing aid; bone-conduction devices suit patients who cannot wear an air-conduction aid, such as those with chronic discharge or canal atresia.

OPERATIVE GUIDES IN THEATRE 6 operations

Myringoplasty and Tympanoplasty
Tympanoplasty

Ossicular Chain Reconstruction
Ossicular reconstruction

Otosclerosis
Stapes surgery

Bone-Anchored Hearing Devices
Bone-anchored hearing

Implantable Middle Ear Devices
Implantable middle ear devices

Cochlear Implantation in Adults
Cochlear implant

4 CHAPTER FOUR Sensorineural loss

Damage at or beyond the cochlea.

4.1 Causes

CAUSES OF SENSORINEURAL LOSS

- CONGENITAL **Genetic, syndromic** intrauterine infection, prematurity, kernicterus
- INFECTION **Viral, meningitis** labyrinthitis, syphilis
- TRAUMA **Head injury, surgery** noise, barotrauma
- TOXIC **Ototoxic drugs**
- DEGENERATIVE **Presbycusis**
- INNER EAR **Ménière's disease** sudden hearing loss, autoimmune disease
- NEURAL **Vestibular schwannoma** multiple sclerosis, CP angle tumours
- SYSTEMIC **Diabetes, thyroid, renal disease** blood disorders, autoimmune disease

4.2 Features

High tones are usually affected more; discrimination is reduced, especially in noise; the loss can exceed 60 dB; there may be recruitment or tinnitus.

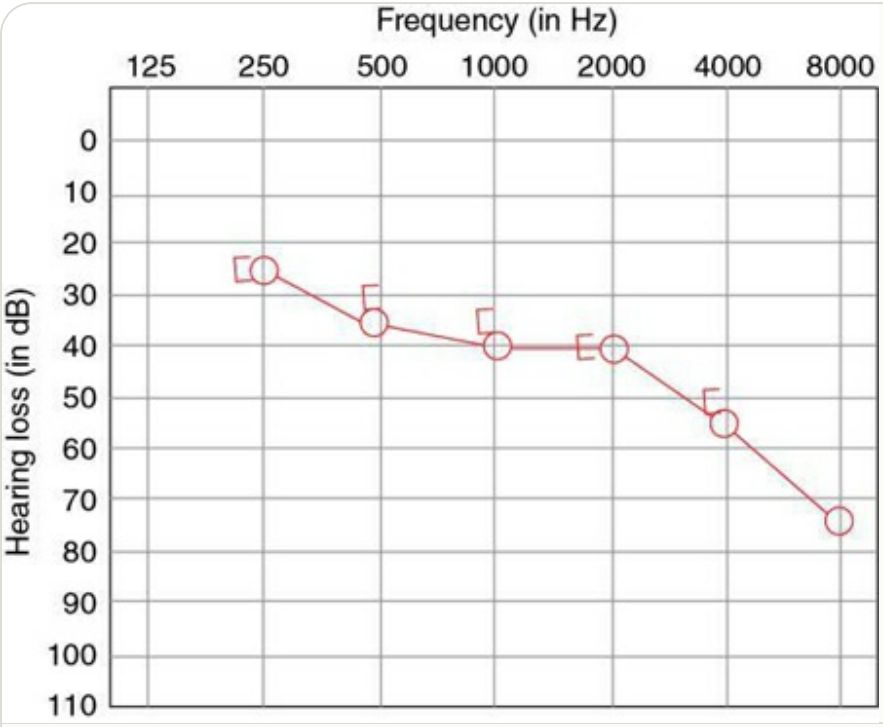


FIGURE 4 Audiogram of a right ear with sensorineural loss: air and bone conduction fall together, with no air-bone gap.

Dhingra, fig. 5.6

4.3 Cochlear or retrocochlear?

COCHLEAR OR RETROCOCHLEAR

TEST	COCHLEAR	RETROCOCHLEAR
Speech discrimination	Reduced, below 90%	Very poor, out of proportion
Roll-over	Absent	Present
Recruitment	Present	Absent
SISI	Over 70%	0–20%
Tone decay	Under 25 dB	Over 25 dB
Stapedial reflex	Present	Absent
Reflex decay	Absent	Present
ABR	Normal I–V	Wave V delayed or absent

Contrast MRI has largely replaced these special audiometric tests for excluding a retrocochlear lesion.

4.4 Management

Treat the cause where possible; protect what remains from noise and ototoxic drugs; then rehabilitate early with hearing aids or implants.

5 CHAPTER FIVE Assessment

Bedside to scanner.

5.1 History

Onset and speed, one or both ears, fluctuation, tinnitus, vertigo, otalgia and discharge, noise and drug exposure, head injury, family history, and general health.

5.2 Examination

Otoscopy under the microscope, tuning forks, and the cranial nerves.

TUNING FORK PATTERNS

TEST	CONDUCTIVE LOSS	SENSORINEURAL LOSS
Rinne	Negative (BC better than AC)	Positive (AC better than BC)
Weber	To the poorer ear	To the better ear
Absolute bone conduction	Normal	Reduced

5.3 Audiometry

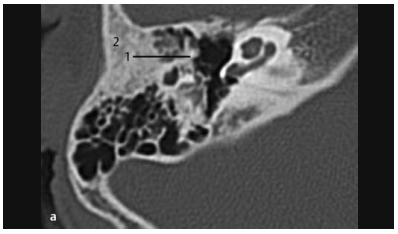
Pure-tone audiogram with masking, speech audiometry, tympanometry and stapedial reflexes; otoacoustic emissions and ABR when needed.

5.4 Further tests by suspicion

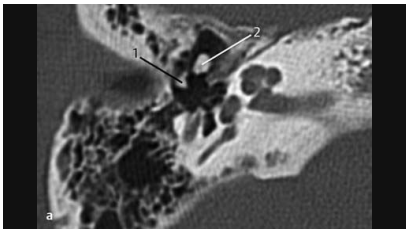
MRI with contrast for asymmetric or retrocochlear loss; CT for conductive loss with a normal drum, trauma or malformation; blood glucose, thyroid and renal function, full blood count and syphilis serology.

IMAGING IN THE APP 3 cases

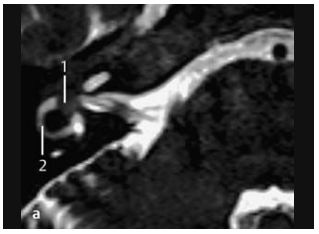
Ossicular fixation and luxation, and vestibular schwannoma.



Fixation of the Ossicular Chain
Bone disease · 3 images



Luxation of the Ossicular Chain
Trauma · 10 images



Vestibular Schwannoma
Tumours · 13 images

5.5 Non-organic hearing loss

CLUES TO NON-ORGANIC LOSS

- Inconsistent repeat audiograms (normal repeats agree within about 5 dB).
- No shadow curve with a claimed one-sided loss.
- Speech reception threshold more than 10 dB better than the pure-tone average.
- A positive Stenger test.
- Stapedial reflexes present despite claimed severe deafness.
- ABR or ASSR near the true threshold.

6 CHAPTER SIX Special situations

When to act fast.

6.1 Sudden sensorineural loss

An emergency: steroids the same day and MRI (see the sudden hearing loss topic).

6.2 Asymmetric sensorineural loss

MRI to exclude a vestibular schwannoma.

6.3 Treatable causes

Syphilis, hypothyroidism, autoimmune inner ear disease, serous labyrinthitis, perilymph fistula and ototoxic drugs.

6.4 Hearing loss in children

Universal **newborn hearing screening** uses automated otoacoustic emissions, followed by automated auditory brainstem responses for babies who do not pass, and all babies who spend time in neonatal intensive care have both. Permanent childhood hearing loss affects about 1 to 2 per 1000 births; about half is genetic

(most often *GJB2*, connexin 26), and congenital cytomegalovirus is the commonest non-genetic cause, tested within the first three weeks of life when antiviral treatment may help. Every child with permanent loss needs an aetiological work-up: imaging, genetics, CMV, ophthalmology, ECG for Jervell and Lange-Nielsen syndrome, and renal assessment when indicated. **Otitis media with effusion** is the commonest cause of acquired hearing loss in children and is managed by watchful waiting for three months, then hearing aids or ventilation tubes if a bilateral loss persists.

6.5 Age-related hearing loss

Presbycusis is a symmetrical high-frequency sensorineural loss. Untreated hearing loss in mid-life is the largest single modifiable risk factor for dementia identified by the Lancet Commission, and hearing aids are associated with reduced cognitive decline in those at risk; this is a strong reason to offer amplification early.

6.6 Implantable hearing

Cochlear implants are offered in the UK to adults and children whose hearing thresholds are 80 dB HL or worse at two or more frequencies between 500 and 4000 Hz in both ears and who gain inadequate benefit from hearing aids, measured by a sentence test score of 50% or less. **Bone-conduction implants**, **middle ear implants** and implants for single-sided deafness extend the options.

6.7 Single-sided deafness

Poor sound localisation and difficulty in noise, even with normal speech elsewhere. Bone-anchored devices, CROS aids or a cochlear implant help; the only hearing ear must be protected.

IN CLINIC

1. Do the tuning forks first: they take seconds.
2. Sudden sensorineural loss goes on the sudden hearing loss pathway today.
3. Asymmetric sensorineural loss needs MRI.
4. Look for treatable causes before fitting an aid.

5 Warn anyone with one hearing ear to protect it.

7 CHAPTER SEVEN Pathway

From complaint to plan.

HEARING LOSS

1 Otoscopy and tuning forks

- → **wax, effusion, perforation or cholesteatoma**
  treat the ear
- → **conductive with a normal drum**
  audiogram, reflexes, CT if surgery considered
- → **sensorineural**
  step 2

2 Speed

- → **sudden, within 72 hours**
  sudden sensorineural loss pathway today
- → **fluctuating with vertigo**
  Ménière's pathway
- → **gradual**
  step 3

3 Symmetry

- → **asymmetric or unilateral**
  MRI of the internal auditory meatus
- → **symmetrical**
  age, noise, drugs, family history, systemic causes

4 Rehabilitation

- → hearing aids
- → bone-conduction device for discharging ears or atresia
- → cochlear implant if thresholds 80 dB or worse with poor aided speech

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Conductive versus sensorineural:** causes, features and fork tests.
- **WHO grades,** old and new.
- **Conductive loss behind an intact drum,** including third-window lesions.
- **Newborn screening** and the work-up of a deaf child.
- **Cochlear implant criteria.**
- **Cochlear versus retrocochlear** tests.
- **Non-organic hearing loss** and how to detect it.
- **Sizes of conductive loss** for common lesions.

8.2 Common pitfalls

COMMON PITFALLS

- Fitting a hearing aid to an asymmetric loss without MRI.
- Missing a sudden loss labelled as a blocked ear.
- Forgetting the treatable causes.

Written for this app from Oxford Textbook of Otolaryngology ch. 5, 11, 12, 83, Oxford Specialist Handbook ch. 3, 14, Oxford Handbook of ENT ch. 3, 4, Dhingra ch. 5. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for

patients.

IMAGING

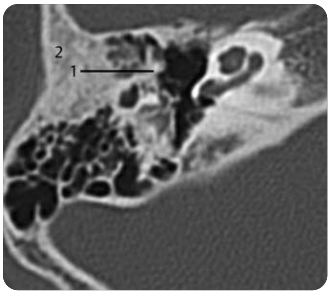
On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · BONE DISEASE

Fixation of the Ossicular Chain

- As mentioned above in differential diagnosis, particular attention must be paid to patients with conductive hearing loss in combination with a normal appearing eardrum and aerated middle ear.
- In case of ankylosis of the ossicular chain, accurate evaluation of thin axial and coronal slices may reveal deformities and/or fixation.



CT, axial. Although the conductive hearing loss might be ascribed to the atresia, the handle of the malleus (1) is fixed to the lateral bony wall of the…

MIDDLE EAR & MASTOID · TRAUMA

Luxation of the Ossicular Chain

IMAGING FINDINGS

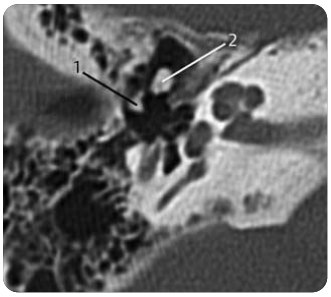
- Separation of the incudostapedial joint; incus rotated or displaced
- Loss of the "ice-cream cone" between malleus head and incus body
- Associated temporal bone fracture through the canal and middle ear

TOP DIFFERENTIALS

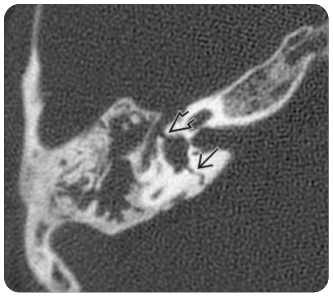
- Ossicular erosion from chronic otitis
- Congenital ossicular malformation
- Partial volume on a single slice

CHECKLIST

- Compare with the other ear on the same slice
- Check facial canal, tegmen and otic capsule



CT, axial. Traumatic luxation with dislocation of the connection between incus (1) and malleus (2) is seen. The typical icecone configuration of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

INNER EAR & IAC · TUMOURS

Vestibular Schwannoma

IMAGING FINDINGS

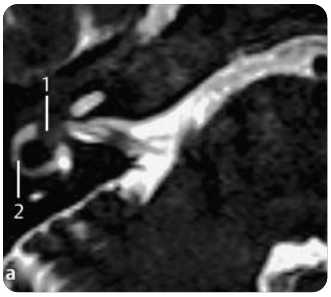
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

TOP DIFFERENTIALS

- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation



MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul…



Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.